

Finite morphisms onto Fano manifolds of large index

(Joint work with Feng Shao)

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ECNU, Algebraic Geometry Seminar, 15th December 2022

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Background & Previous results

- Throughout this talk, we work over $\mathbb{C}$.
- Let X be a projective variety. The **Picard number** of X is defined as $\rho(X) := \dim_{\mathbb{R}} \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$ where $\text{NS}(X) = \text{Pic}(X)/\text{Pic}^{\circ}(X)$ is the **Néron-Severi group** of X .

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- A smooth projective variety X is said to be **Fano**, if the anti-canonical divisor $-K_X$ is ample Cartier.
- Let X be a smooth Fano projective variety. The **(Fano) index** of X is defined as the maximal integer $i(X)$ such that $-K_X \sim i(X)H$ for some integral ample Cartier divisor H (which we call the **fundamental divisor** of X when $\rho(X) = 1$).

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- A Fano manifold X is said to be **del Pezzo** if $i(X) = \dim(X) - 1$. In this case, define $d := H^{\dim(X)}$ as the **degree** of X .
- Let X be a Fano manifold of $\rho(X) = 1$. A smooth rational curve ℓ on X is said to be a **line**, if $H \cdot \ell = 1$ where H is the fundamental divisor.

Background & Previous results

- Let $f : Y \rightarrow X$ be a finite surjective morphism between smooth projective varieties X and Y of the same dimension.

Question (Rough version)

Are there only finitely many such maps? Does there exist a number N which depends only on X and Y such that all of the finite morphisms $f : Y \rightarrow X$ have the degree no more than N ?

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- (A special case) If $\rho(X) = 1$ and K_X is big, ramification divisor formula implies that the degree is always bounded.
- If X is an abelian variety or a projective space, the above questions clearly have negative answer since such X have rich endomorphisms.

Background & Previous results

Conjecture (Amerik, 97; Van de Ven's conjecture)

Let $f : Y \rightarrow X$ be a finite morphism between smooth projective varieties of $\dim n \geq 2$ and of the second Betti number 1. Suppose $X \not\cong \mathbb{P}^n$. Then $\deg(f)$ is bounded in terms of the discrete invariants of Y and X .

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The conjecture is known in the following cases:

- (Kobayashi-Ochiai, 75) X is of general type.
- (Amerik-Rovinsky-Van de Ven, 99) X is strict Calabi-Yau.
- (Amerik, 97; Amerik-Rovinsky-Van de Ven, 99) $\dim(X) \leq 3$.
- (Amerik, 07) X is a smooth quadric hypersurface of $\dim(X) \geq 3$.

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Main question (Hwang-Mok, 03)

Let X and Y be Fano manifolds of $\dim = n$ and of $\rho = 1$. Suppose that $X \not\cong \mathbb{P}^n$. Is there a number N depending only on X and Y such that every finite morphism $Y \rightarrow X$ has the degree no more than N ?

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- (Amerik-Rovinsky-Van de Ven, 99), (Hwang-Mok, 03) Fano 3-folds
- (Paranjape-Srinivas, 89), (Beauville, 01) hypersurfaces of $\mathbb{P}^n$
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Theorem (Hwang-Mok, 99; Lazarsfeld's problem)

Any surj. holo. map from a rational homogeneous manifold of $\rho = 1$ onto a projective manifold different from the projective space, is biholomorphic.

Background & Previous results

As a higher Picard number analogue, Fakhruddin asked the following.

Question (Fakhruddin, 03); cf. (Meng-Z., 20)

Let X be a rationally connected smooth projective variety. Suppose X admits a polarized or an int-amplified endomorphism. Is X a toric variety?

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- A surjective endomorphism $f : X \rightarrow X$ of a projective variety X is **polarized** if $f^*H \sim qH$ for some ample Cartier divisor H on X and integer $q > 1$ ($\Leftrightarrow f^*|_{N^1(X)}$ diagonalizable with e.vs modulus q), and **int-amplified** if $f^*H - H$ is ample for some ample Cartier divisor H on X ($\Leftrightarrow$ e.vs of $f^*|_{N^1(X)}$ modulus > 1); (Meng-Zhang, 18), (Meng, 20).

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- A normal variety X of dim n is a **toric** variety if X contains a big torus $T = (\mathbb{C}^*)^n$ as an (affine) open dense subset such that the natural multiplication action of T on itself extends to an action on X .
- Toric varieties having many interesting dynamical symmetries and morphisms to toric varieties do not satisfy any boundedness property.

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This question has a positive answer in the following cases:

- (Nakayama, 02) $\dim(X) = 2$.
- (Hwang-Nakayama, 11), (Meng-Zhang, 19), (Meng-Z., 20) The ramification is totally invariant. Recall that a closed subset $Z \subseteq X$ is f^{-1} -invariant or f -totally invariant for a surjective endomorphism $f : X \rightarrow X$ if $f^{-1}(Z) = Z$.
- (Meng-Zhang-Z., 22) X is a Fano threefold.
- (Jia-Z., 22) X is a Fano fourfold admitting a conic bundle structure.

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We shall study the above questions/conjectures for Fano manifolds with large Fano index. Today, we will mainly focus on the case $\rho(X) = 1$.

Background & Previous results – del Pezzo manifolds

(Fujita, Iskovskikh) Let (X, H) be an n -dim del Pezzo manifold of degree $d = H^n$. Assume that $n \geq 3$. Then X is one of the following.

- $d = 1$ and X is a smooth weighted hypersurface of degree 6 in $\mathbb{P}(3, 2, 1^{\oplus n})$.
- $d = 2$ and X is a smooth weighted hypersurface of degree 4 in $\mathbb{P}(2, 1^{\oplus(n+1)})$.
- $d = 3$ and X is a smooth cubic hypersurface in $\mathbb{P}^{n+1} = \mathbb{P}(1^{\oplus(n+2)})$.
- $d = 4$ and X is a smooth complete intersection of two smooth quadrics in $\mathbb{P}^{n+2}$.
- $d = 5$, $n \leq 6$ and $X \subseteq \mathbb{P}^{n+3}$ is a linear section of the Grassmannian $\mathrm{Gr}(2, 5) \subseteq \mathbb{P}^9$ under the Plücker embedding.
- $d = 6$, $n \leq 4$ and X is $\mathbb{P}^2 \times \mathbb{P}^2$, $\mathbb{P}(T_{\mathbb{P}^2})$ or $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$.
- $d = 7$ and X is a blow-up of a point on $\mathbb{P}^3$.

Moreover, $\rho(X) = 1 \Leftrightarrow d \leq 5$, and H is very ample $\Leftrightarrow d \geq 3$.

Main results & Proofs

Theorem A (Shao-Z., 22)

Let X be a del Pezzo manifold of Picard number 1. Then X does not admit any non-isomorphic surjective endomorphism.

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Theorem (To deal with del Pezzo manifolds of $\deg \leq 3$)

Let X be a smooth weighted hypersur of dim $n \geq 2$ and of deg d in $\mathbb{P}(a_0, a_1, \dots, a_{n+1})$ with $a_0 \geq a_1 \geq a_i = 1$ for each $i \geq 2$. If $(a_0, a_1) = 1$ and $d \geq a_0 + a_1 + 1$, then for any smooth proj variety Y of $\dim(Y) = n$ and $\rho(Y) = 1$, the deg of any finite morphism $f : Y \rightarrow X$ is bounded.

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- Let H be a suitable very ample divisor. We extend a Hurwitz type formula in (Amerik-Rovinsky-Van de Ven, 99):
 $\deg(f)c_n(\Omega_X(H)) \leq c_n(\Omega_Y(f^*H)).$
- By residue theorem, we have $c_n(\Omega_X(uH)) > u^n H^n$ for some $u > 0$.
- Get the boundedness property by the expression of top Chern classes.

Main results & Proofs

A direct consequence of the previous theorem is the following corollary.

Corollary; cf. (Beauville, 01), (Amerik, 07)

Let $X \subseteq \mathbb{P}^{n+1}$ be a smooth hypersur of deg d . If either $d \geq 3$, or $d = 2$ and $n \geq 3$, then any finite surjective morphism $Y \rightarrow X$ from a smooth proj variety Y of $\rho(Y) = 1$ has bounded degree.

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Question

Can we show the boundedness property for morphisms onto weighted hypersurfaces/complete intersections in a more general setting?

- So far, all of the weighted projective spaces we considered have only isolated singularities.
- This is enough for our purpose to deal with del Pezzo manifolds. The next step in this direction is to consider the case when all the weights a_i are mutually coprime (well-formed weighted hypersurfaces).

Main results & Proofs

Theorem (To deal with del Pezzo manifolds of $\deg = 4$)

Let X be a del Pezzo manifold of degree 4 and of $\dim n \geq 3$. For any smooth proj variety Y of $\dim n$ and of Picard number 1, there is $N > 0$ such that the degree of any finite morphism $Y \rightarrow X$ is no more than N .

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- (Reid, 73) After a suitable change of basis, $X \subseteq \mathbb{P}^{n+2}$ is given by the complete intersection of two smooth quadrics Q_1 and Q_2

$$Q_1 : \sum_{i=0}^{n+2} x_i^2 = 0, \quad Q_2 : \sum_{i=0}^{n+2} \lambda_i x_i^2 = 0$$

with $\lambda_i \neq \lambda_j$ if $i \neq j$. Then the projection $\pi_i : X \rightarrow Q^n$ is a double cover where Q^n is a smooth quadric defined by $\sum_{j \neq i} (\lambda_j - \lambda_i) x_j^2 = 0$.

- (Amerik, 07) Finite surjective morphisms from a smooth projective variety of $\rho = 1$ onto a smooth quadric have the degrees bounded.

Main results & Proofs

Theorem (To deal with del Pezzo fourfolds of $\deg = 5$)

Let X be a del Pezzo fourfold of degree 5. Then for any smooth projective fourfold Y with $b_2(Y) = 1$, there is a positive number N such that any finite morphism $Y \rightarrow X$ has the degree no more than N .

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- We extend (Amerik, 97) on quintic del Pezzo threefolds to fourfolds.
- The reason that we can extend: the covering locus of the lines of “negative type” is a prime divisor (of codimension one) so that we can apply the induction and the relative normal bundle sequence would split (we skip the detailed technical proofs herein).
- This method does not work any more when $\dim(X) = 5$ or 6 since the lines are much more “positive” (the surjection $T_X|_{\ell} \rightarrow \mathcal{N}_{\ell/X}$).

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- End of Proof of Theorem A: del Pezzo manifolds of degree 5 and of $\dim \geq 4$ are quasi-homogeneous cf. e.g. (Fu-Montero, 19), so we can apply (Hwang-Nakayama, 11) and (Paranjape-Srinivas, 89).

Main results & Proofs

Theorem B (Shao-Z., 22)

Let X be a smooth Fano 4-fold of $\rho(X) = 1$ and $i(X) > 1$, and $X \not\cong \mathbb{P}^4$. Then for any smooth Fano 4-fold Y of $\rho(Y) = 1$, there is a number N such that any surjective morphism $Y \rightarrow X$ has the degree no more than N . In particular, X does not admit any non-isom surj endo.

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- (Kobayashi-Ochiai, 73; cf. Cho-Miyaoka-Shepherd-Barron, 02) If $i(X) = 5$, then $X \cong \mathbb{P}^4$.
- If $i(X) = 4$, then X is a quadric (Miyaoka, 04; Dedieu-Höring, 17) and Theorem B follows from (Amerik, 07).
- If $i(X) = 3$, then X is a del Pezzo fourfold and Theorem B follows from Proof of Theorem A.
- If $i(X) = 2$, then by a standard calculation on the normal bundle of lines, we can show X has trivial VMRT ($p = 0$) along a general point, in particular, Theorem B follows from (Hwang-Mok, 03).

$$T_X|_\ell = \mathcal{O}(2) \oplus \mathcal{O}(1)^p \oplus \mathcal{O}^{3-p}$$

Main results & Proofs

Theorem C (Shao-Z., 22)

Let X be a Fano manifold of Picard number ≥ 2 . If $i(X) \geq \lfloor \frac{\dim(X)+1}{2} \rfloor$, then X is toric if and only if X admits an int-amplified endomorphism.

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Together with the results on the case $\rho(X) = 1$, we have the following corollary. The case of Mukai manifolds of $\rho(X) = 1$ hasn't been finished.

Corollary D (Shao-Z., 22)

Let X be either a smooth Fano fourfold of index > 1 , or a del Pezzo manifold. If X admits an int-amplified endomorphism, then X is toric.

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- We used the classification for del Pezzo manifolds, Mukai manifolds and manifolds with middle Fano index.
- Most techniques we used are inspired from (Meng-Zhang-Z., 22) by endo-descending. For example, if X admits a Mori contr to a smooth quadric of $\dim n \geq 3$, then X does not admit any int-amplified endo.

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- In (JZ22): if a Fano 4-fold admits an elementary conic bundle, then the same result follows; however, we don't know whether a Fano contr. τ of rel dim 1 from a Fano 4-fold is equi-dim under the dynamical assumpt. so that τ is a conic bundle (Ando, 85).

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- In (SZ22): this expectation is hopefully true ($i(X) > 1$ ✓).
- In (JZ22): if a Fano 4-fold X admits a singular conic bundle (factor through blow-ups), then there is a rational curve ℓ with $-K_X \cdot \ell = 1$; thus X has index 1. However, a Fano 4-fold of index 1 does not necessarily admit a conic bundle, so the general case is widely open.

Further discussions & Open questions (Back to $\rho(X) = 1$)

To complement the boundness problem for finite morphisms onto Fano fourfolds, we are left to consider the index 1 case.

Question

*Let X be a smooth Fano 4-fold of $\rho(X) = 1$ and $i(X) = 1$. Suppose the variety of minimal rational tangents $\mathcal{C}_x$ along a general point $x \in X$ is **1-dim**. For any smooth Fano 4-fold Y of $\rho(Y) = 1$, is there a number N such that every surj morphism $Y \rightarrow X$ has the degree no more than N ?*

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- Let X be a uniruled projective manifold and $\text{RatCurves}^n(X)$ the normalized space of rational curves on X .
- An irreducible component $\mathcal{K}$ of $\text{RatCurves}^n(X)$ is called a **family of rational curves**.
- Let $e : \mathcal{U} \rightarrow X$ be the normalization of the universal family over $\mathcal{K}$ and let $q : \mathcal{U} \rightarrow \mathcal{K}$ be the natural $\mathbb{P}^1$ -fibration. We denote by $\mathcal{K}_x$ the normalization of the set $q(e^{-1}(x))$ for some point $x \in e(\mathcal{U})$.

Further discussions & Open questions

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{e} & X \\ q \downarrow & & \\ \mathcal{K} & & \end{array}$$

- The family $\mathcal{K}$ is **dominating** if e is dominant and $\mathcal{K}$ is **locally unsplit** if $\mathcal{K}_x$ is proper for a general $x \in e(U)$.
- A general $[\ell]$ in a locally unsplit dominating family $\mathcal{K}$ is **standard**, i.e., the pullback of T_X to the normalization $f : \mathbb{P}^1 \rightarrow \ell$ splits as follows:

$$f^* T_X \cong \mathcal{O}_{\mathbb{P}^1}(2) \oplus \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus p} \oplus \mathcal{O}_{\mathbb{P}^1}^{\oplus (n-p-1)}$$

where $n = \dim(X)$ and $p = \dim(\mathcal{K}_x)$ for a general point $x \in X$.

- Associated with a locally unsplit dominating family $\mathcal{K}$ and a general point $x \in X$, the tangent map $\tau_x : \mathcal{K}_x \dashrightarrow \mathbb{P}(\Omega_{X,x})$ sends a rational curve that is smooth at x to its tangent direction.

Further discussions & Open questions

- The closure $\mathcal{C}_x$ of the image of τ_x is called the **variety of minimal rational tangents** (VMRT) at the point x associated to $\mathcal{K}$. The tangent map $\tau_x : \mathcal{K}_x \dashrightarrow \mathcal{C}_x$ is in fact the normalization morphism (Kebekus, 02) and (Hwang-Mok, 04), and $\dim(\mathcal{C}_x) = \dim(\mathcal{K}_x) = p$.

Remark (Reduction on the case $i(X) = 1$)

Let X be a Fano 4-fold of $\rho(X) = 1$ and $i(X) = 1$. Let $\mathcal{K}$ be a dominating family of m.r.c.. Given a standard ℓ of $\mathcal{K}$, we have

$$T_X|_{\ell} \cong \mathcal{O}_{\ell}(2) \oplus \mathcal{O}_{\ell}(1)^{\oplus p} \oplus \mathcal{O}_{\ell}^{\oplus(3-p)}.$$

Then $0 \leq p \leq 3$. If $p = 0$, then VMRT is 0-dim, so the boundedness property follows from (Hwang-Mok, 03). If $p = 2$, then $-K_X \cdot \ell = 4$ and X is a quadric by (Miyaoka, 04). If $p = 3$, then $-K_X \cdot \ell = 5$ and $X \cong \mathbb{P}^4$. We are left to the case $p = 1$.

Another difficulty is that we don't know whether such X has/is covered by lines with respect to some embedding.

Further discussions & Open questions (Singular quadrics)

Question (A generalization of smooth quadrics; cf. Amerik, 07)

Let X be a (possibly singular) quadric hypersurface in $\mathbb{P}^n$ and let Y be a normal projective variety of Picard number 1. Does every finite surjective morphism $Y \rightarrow X$ have bounded degree?

- Motivation: study rational homo variety admitting a finite morphism onto a singular quadric so that the above question can help study the boundedness problem of morphisms onto such rational homo varieties.
- Motivated by (Nakayama-Zhang, 10) and (Yanis, 21), we showed that a singular quadric X admits a non-isom surj endo $\Leftrightarrow$ the rank of X is no more than 3, i.e., after a suitable linear map, X is defined by $\sum_{i=0}^k x_i^2 = 0$ where $k \leq 3$.
- We are not sure whether the quadratic equations for Grassmannians can (dis)prove the existence of morphisms onto singular quadrics.

Further discussions & Open questions

Question (Lifting and descending problem)

Let X be a linear section of a rational homogeneous variety. Let Y be a smooth hyperplane section of X which is of Picard number 1. Given a surjective endomorphism f on Y , can we lift it to X ?

- When X is the projective space and Y is a closed subvar with some good embedding property, (Fakhruddin, 03) confirms the existence of the lifting. When X is some other rational homogeneous variety, we knows very few things.
- (Manivel-Benedetti, 22) dealt with the case when $\deg(f) = 1$.
- A positive answer to the above question will almost conclude the case of Mukai manifolds of $\rho = 1$, i.e., Mukai manifolds of $\rho = 1$ do not admit any non-isomorphic surjective endomorphism (the genus $g := \frac{1}{2}H^n + 1$ plays a role). This is our work in progress.

Thank you for your attention!